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STAND-UP

Standup Meeting



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Share yesterday and choose one next action

1

What happened?

Describe one event or result.

2

What did it mean?

Share a feeling, an interpretation and your next action.

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REVIEW

Review the Defect Investigation Homework



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Review the submitted diagnosis and repair

1

Show your submission

Explain the diagnosis, code change and verification.

2

Give specific feedback

Point to a code change or test result and explain your comment.

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BREAK

Take a short break



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LECTURE

Explain Unit Testing and Mockito



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Four prices for the same 3 kg parcel

Local starts at 1000 cents and remote at 2000. Add 300 cents per kg above 2 kg and 1500 cents for express.

Zone · Local

Service

Standard	Express
1300 cents	**2800 cents**

Zone · Remote

Service

Standard	Express
2300 cents	**3800 cents**

Local standard · 1000 base + 300 for one extra kg = 1300 cents

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Check 1300 cents through the API

QuotePriceComparisonTest · checkpoint/day10-testing

```
@Test
void pricesLocalStandardAt3Kg() throws Exception {
String body = """
    {"weight_kg":3,"zone":"local","service":"standard"}
    """;
mvc.perform(post("/quotes").contentType(MediaType.APPLICATION_JSON).content(body))
    .andExpect(status().isCreated())
    .andExpect(jsonPath("$.amount_cents").value(1300));
}
```

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A crash-test dummy stands in for a person

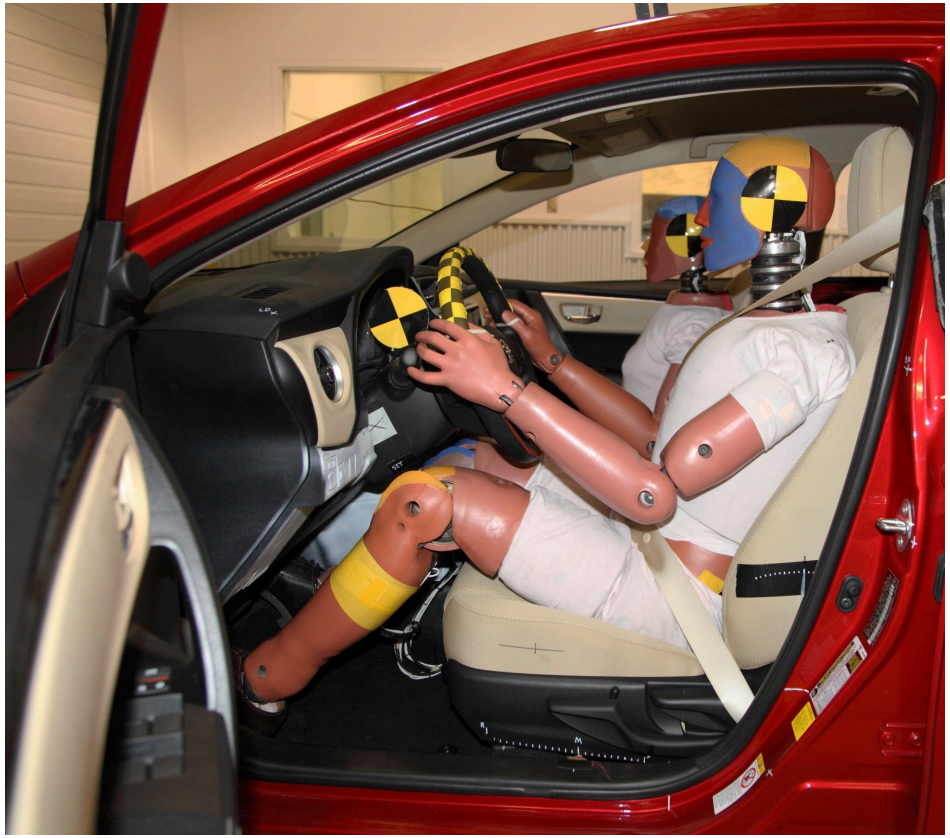


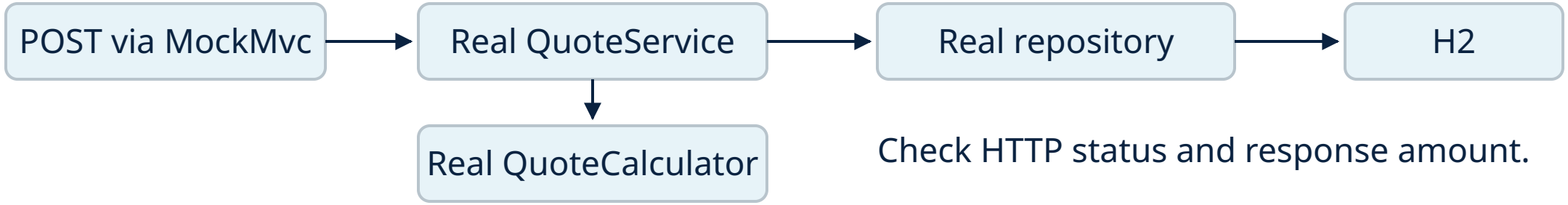
Photo from NHTSA crash test 9984 · Public domain (U.S.)

A **test double** takes the place of a real collaborator during a test.

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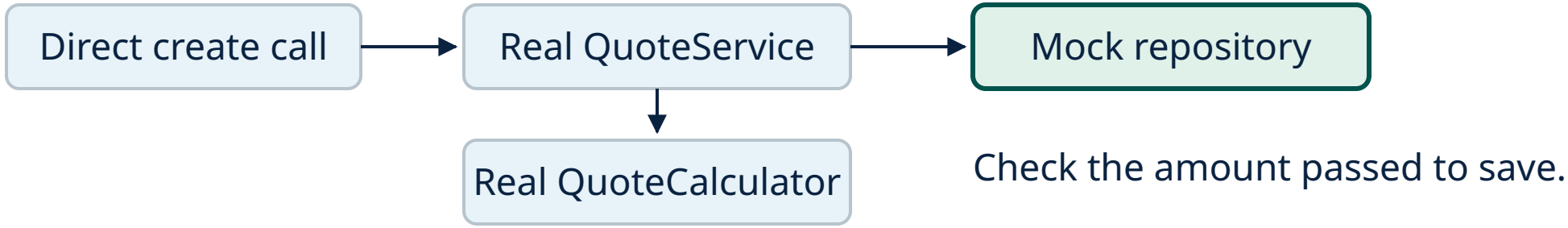
Keep the pricing code and replace the repository

API test



Check HTTP status and response amount.

Service test



Check the amount passed to save.

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Check the amount passed to save

Service test · completed example for S17

```
@Test
void pricesLocalStandardAt3Kg() {
QuoteRepository repository = mock(QuoteRepository.class);
QuoteService service =
    new QuoteService(repository, new QuoteCalculator());
var input = new ParcelInput(3, "local", "standard");

service.create(input);
verify(repository).save(null, input, 1300);
}
```

QuoteService.create · return statement

```
return repository.save(null, input, calculator.calculate(input));
```

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Set a return value or check a call

Stub a return value Set what the repository returns when the Service reads an existing quote.

```
when(repository.find(17L)).thenReturn(Optional.of(existingQuote));
```

Verify a call Check the arguments the Service passed when it created the quote.

```
verify(repository).save(null, input, 1300);
```

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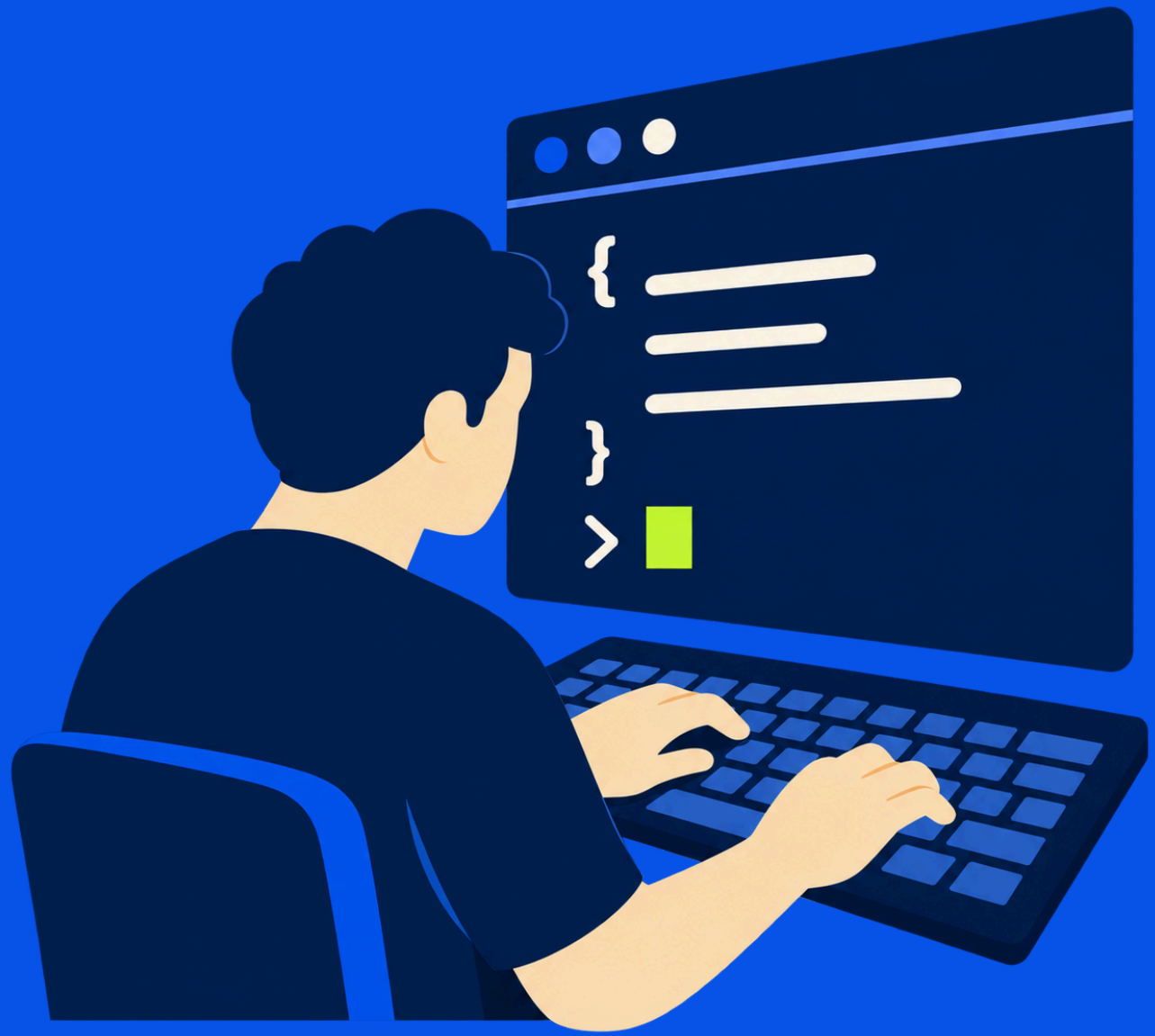
What does a passing Service test tell us?

Imagine the bug	The Service passes 1300 cents to save, but the POST response has no ID.
Predict the result	Would the Service test fail? Which check would catch the missing ID?

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DEMO

Demonstrate Moving One Pricing Test to the Service Layer



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Move one pricing example to a Service test

1

Find the API test

3 kg, local, standard costs 1300 cents.

2

Write and run the Service test

Create QuoteServiceCreatePriceTest. Verify that the mock repository receives 1300 cents.

3

Remove the matching API test

Keep CreateApiTest for the real POST, generated ID and database write.

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Move API Pricing Examples to Service Unit Tests



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Move three more pricing checks at 3 kg

1

Find the API examples

Local express: 2800 cents.
Remote standard: 2300 cents.
Remote express: 3800 cents.

2

Write three methods in the same class

Continue in QuoteServiceCreatePriceTest. Verify the amount passed to the mock repository for each combination.

3

Remove the three matching API tests

Delete the empty pricing test class. Keep the HTTP, persistence and rejection tests, then rerun the suite.

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BREAK

Take a short break

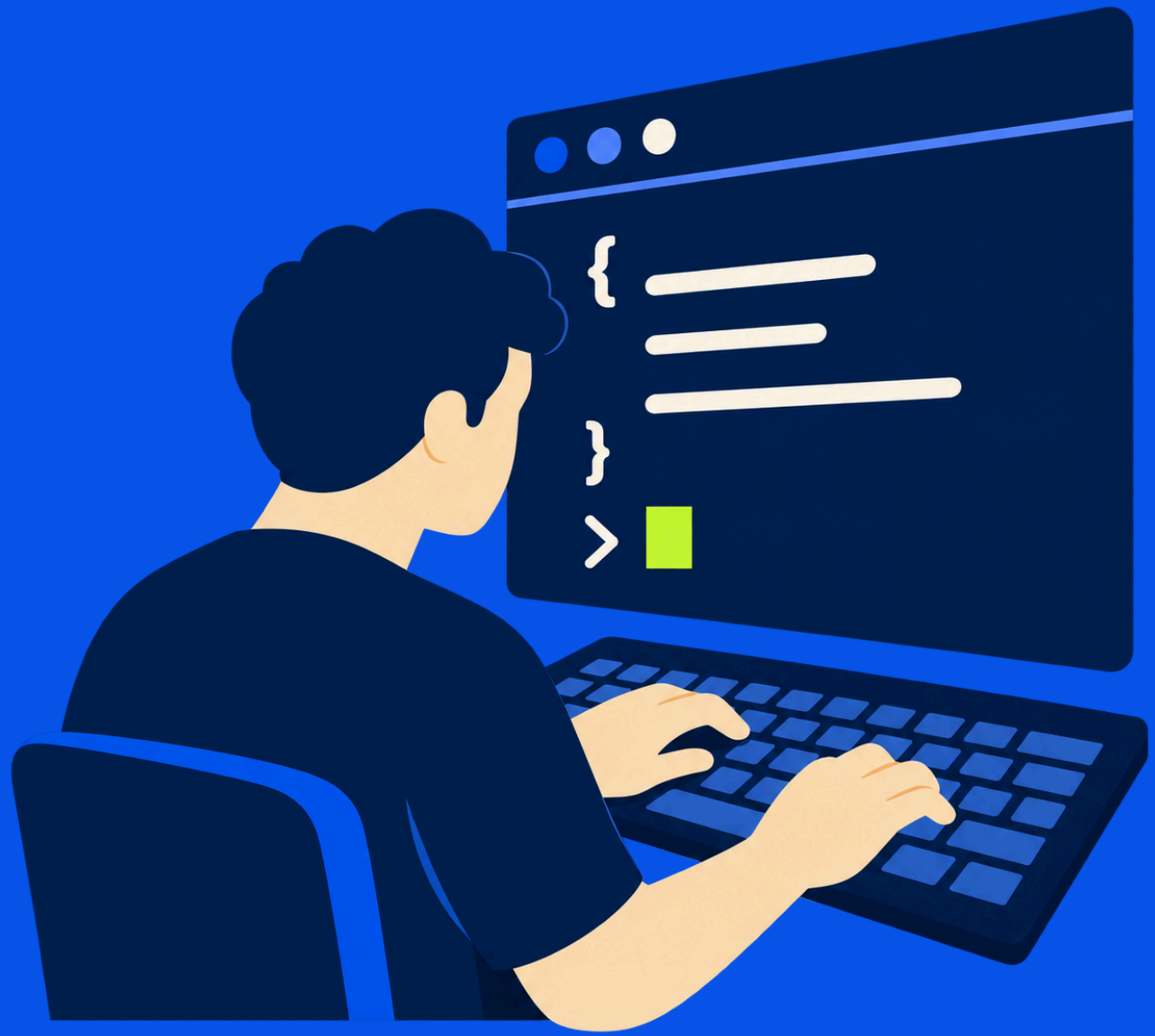


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DEMO

Demonstrate a Browser End-to-End Regression



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How a test drives a real browser

Playwright Test	Runs the test and gives it a browser tab as `page`.
↓	
Browser tab	Opens the app, fills the **Weight** field and clicks **Create**.
↓	
Page result	Playwright checks the page for the **Order created** message.

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Create an order and check the message

Small example · assumes the order app is running on port 3000

```
import { test, expect } from '@playwright/test';

test('A user can create an order', async ({ page }) => {
  await page.goto('http://localhost:3000');

  await page.getByLabel('Weight').fill('2');
  await page.getByRole('button', { name: 'Create' }).click();

  await expect(page.getByText('Order created')).toBeVisible();
});
```

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Read the test in three parts

1

Open the app

`test(...)` defines the test. Playwright gives it a fresh `page`. `goto` opens the app there.

2

Find and act

A locator describes what to find. `getByLabel` targets the field. `getByRole` targets the button. `click()` waits for it to be ready.

3

Check the result

`getByText` locates the message. `toBeVisible()` checks again until the message is visible or the assertion times out.

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Listen for the quote response before clicking

In the quote demo, we also need the ID returned by the server.

```
const saved = page.waitForResponse(response =>
  response.request().method() === 'POST' &&
  response.url().endsWith('/api/quotes'));
await page.getByRole('button', { name: 'Save quote' }).click();
const response = await saved;
const { id } = await response.json();
```

Start listening first so a fast response is captured. Then use its ID to check the matching row on the page.

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Follow the same 5900-cent quote through reload

1

Save the quote

Create a 10 kg remote express quote. Read the ID from the POST response and locate the matching row.

2

Reload the list

Check express and 5900 cents on the same row. Open the details for that ID.

3

Repeat in isolation

Reload uses the same backend and database. An independent run starts with a fresh database.

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Complete and Repeat One Browser Regression



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Check that the saved amount survives a reload

1

Continue in the same checkout

Complete the scaffold for 3 kg, local, standard. Expect 1300 cents.

2

Check the same quote

Use the POST ID to find the row before and after reload. Verify 1300 cents both times. Open details for the same ID.

3

Run twice

Confirm both lists start empty. Compare the H2 paths in the two environment attachments.

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SHOWCASE

Review the Browser Testing Practice



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Review the browser tests and their run evidence

Group 1 · Reload	Show the POST ID and the assertion that would catch 1300 cents changing to 13 after reload.
Group 2 · Isolation	Show the empty list and database path for each run, or show the actual blocker.
Apply the feedback	Choose one assertion, wait or piece of run evidence to check in your own test.

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LECTURE

Explain the Test Pyramid and Testing Quadrants



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What catches the wrong amount after refresh?

Expected quote	3 kg, local, standard → 1300 cents
Browser path	Save the quote, keep its ID, find its row, then refresh.

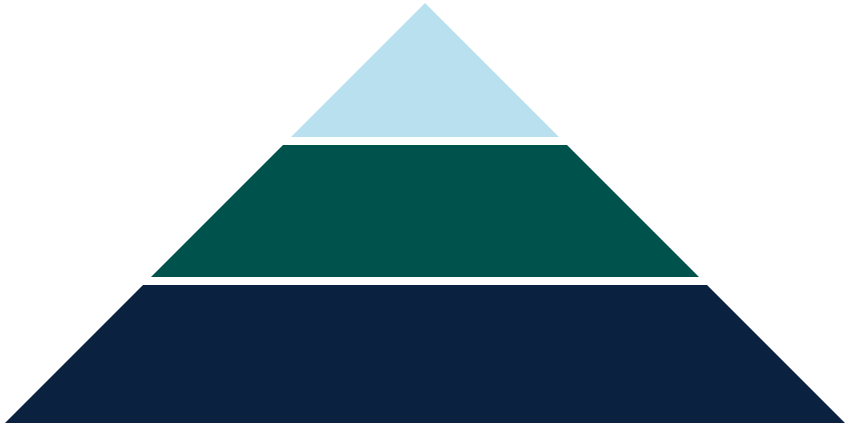
Suppose the amount shows 13 after refresh. Which assertion catches it?

Use the ID from the create response. After refresh, check that the same row shows **standard · 1300 cents**.

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Where should a new pricing test go?

Service	Check the price passed to a mock repository.
API + database	Check the stored data and what happens when a request is rejected.
Browser	Save, refresh, and check the amount in the same row.



Browser	Protect the save and refresh path.
API + database	Protect storage and rejected requests.
Service	Cover pricing boundaries with fewer dependencies.

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Can a first-time user find their quote?

Suppose Playwright passes. After refresh, the amount for the same quote ID is correct.

What can we learn from this result? What would you check with a new user?

Run the script. Does the expected behavior still work?

Watch a new user. How do they recognize the right quote?

	Support the team	Evaluate the product
Business-facing	**Q2 · Automated and manual** Story, prototype or simulation, regression, and end-to-end tests	**Q3 · Manual** Exploratory testing, user acceptance testing, and usability testing
Technology-facing	**Q1 · Automated** Unit, component integration, and API tests	**Q4 · Tool-assisted** Performance, reliability, security, and compatibility testing

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What does a green run tell the team?

Service pricing	API and database	Browser save and refresh

After a small change is merged into main, which checks would you run often?

If only the Service check runs and passes, what can you tell your team?

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BREAK

Lunch



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PRACTICE

Experience Integration Feedback



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The Integration Game · Build and connect a railway

1

Build your section

Use the A4 map. Connect the marked sides through your station and avoid obstacles.

2

Connect the railway

Follow the instructions for when to put finished sections on the shared wall. Check that the routes meet.

3

Keep the evidence

Record first feedback, mismatches, rework, checking costs and unfinished work. Either round may finish or remain incomplete.

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Reflect · When did integration give you useful feedback?

Observe	When did a mismatch become visible? What was already built on the earlier assumption?
Explain	What needed rework? What did repeated integration and checking cost?
Transfer carefully	Use actual records. A manual paper check is not test automation or a software productivity experiment.

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LECTURE

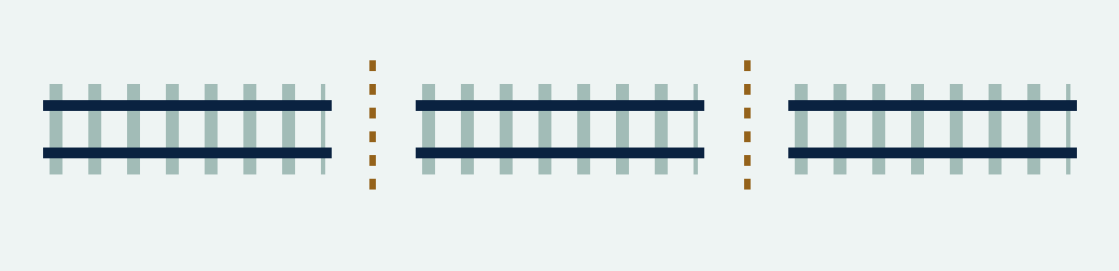
Explain Continuous Integration



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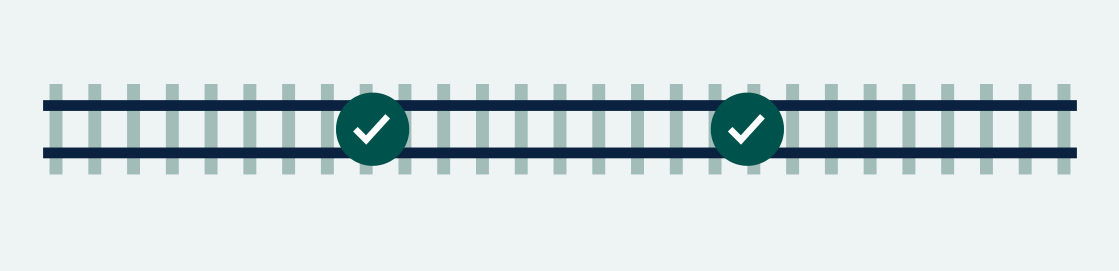
When could you check the seam?

Zhuhai · Connect at the end



Draw separately. Put all sections on the wall together.

Hong Kong · Connect as you go

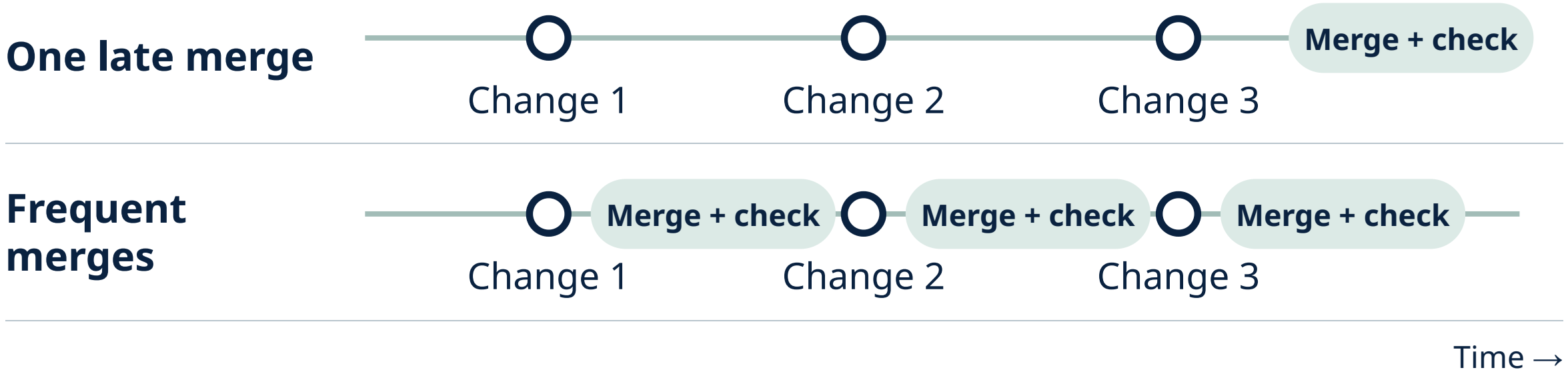


Put each finished section up. Check the seams you can reach.

Pick one seam from your game. When could you check it?

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How long do changes wait for feedback?



If a check fails, how many changes might you need to inspect?

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What is continuous integration?

Integrate small changes into a shared mainline frequently. **Automatically build and test the combined version.**



Track section
Small change

Shared wall
Merge into main

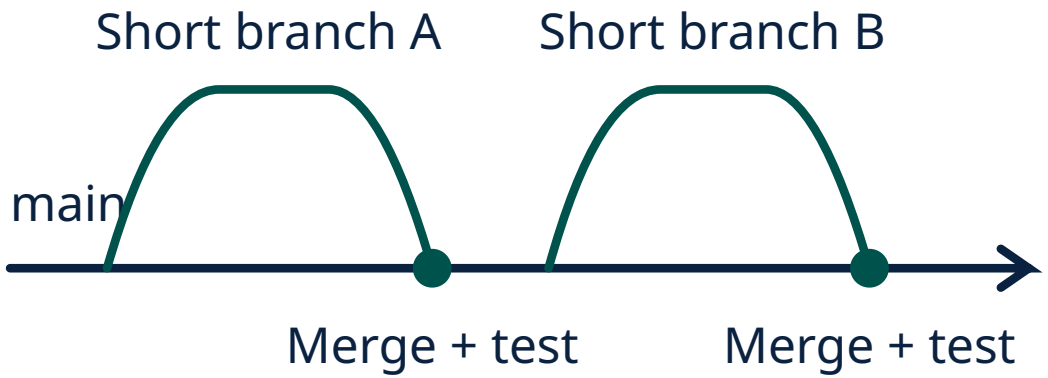
Check the seam
Build and test

Repair and recheck
Fix and verify

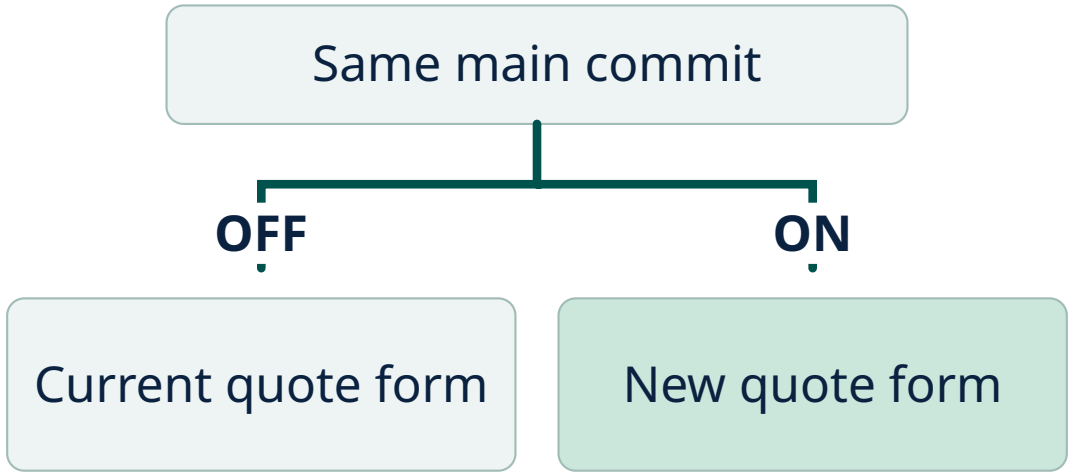
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Keep main ready for the next change

Trunk-based development



Feature toggle



If a check fails, fix or revert the change. Then run the checks again.

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Do you need CI when you work alone?

One developer · main



Saved quotes reload

✓ **Passed**

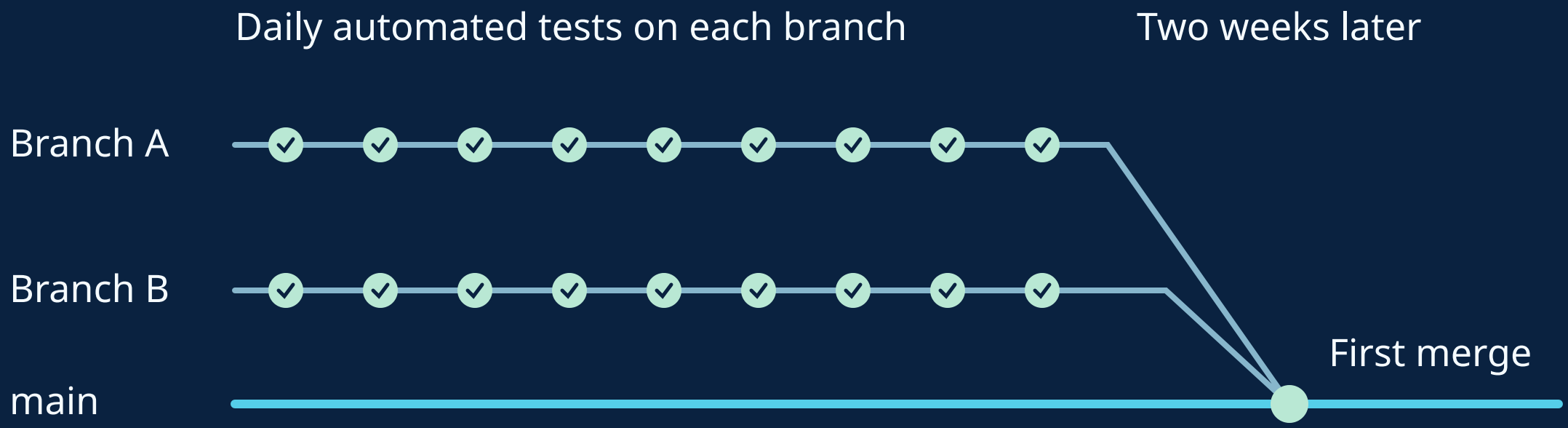
Update a dependency

Do quotes still reload?

Run the same check

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Is this continuous integration?



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Takeaway

Integrate Early.
Integrate Often.
Keep It Working.

Continuous Integration is a team discipline, enabled by automation.

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BREAK

Take a short break



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SHOWCASE

Teach Group Topics and Connect Engineering Evidence



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Teach a design problem, example and tradeoff

1

Explain

Use your prepared group slides: a concrete problem, code, collaboration and a tradeoff. Explain when a simpler design is enough.

2

Involve everyone

Each learner explains part of the topic or answers a question. Ask the audience to judge an example.

3

Feedback and voting

Receive instructor feedback after each group. Use the final 10 minutes to vote using the announced criteria.

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BREAK

Take a short break



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SHOWCASE

Continue group teaching and instructor feedback



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BREAK

Take a short break



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RETRO

Review and Improve Group Collaboration



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Review how your group worked and choose one improvement

1

Look back

Revisit the Day 5 action. What did you try, and what did you observe?

2

Discuss real events

Use Week 2 collaboration, including preparation and group teaching. Let every member contribute.

3

Choose one action

Agree what to try next, who will remind the group and what you will observe after the next group activity.

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LECTURE

Brief the Homework Test Workflow



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Homework · Run Maven tests in GitHub Actions

1

Start from your classroom project

Use the project where you moved pricing tests from the API layer to the Service layer. Confirm the local Maven tests pass. You may use AI to draft and troubleshoot the YAML.

2

Run the full Maven suite on push

Use one job on ubuntu-24.04. Check out the commit, set up Java 21, then run ./mvnw test.

3

Submit your workflow and ORID

Push .github/workflows/test.yml and records/orid.md. In Actions, check that the full test suite and workflow pass.

Example run · push · commit a13f924 · Tests passed · Workflow passed

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